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Progression from impaired glucose tolerance to type 2 diabetes in obese children and adolescents: a 3–6-year cohort study in southern Thailand

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Abstract

Background: Childhood obesity is associated with abnormal glucose metabolism and type 2 diabetes mellitus (T2DM). This study evaluated the prevalence of abnormal glucose metabolism in asymptomatic obese children and adolescents, and determined the percentage of T2DM development after 3–6 years of follow-up.
Methods: During 2007–2013, 177 obese children and adolescents who had normal fasting plasma glucose (FPG < 100 mg/dL) were given an oral glucose tolerance test (OGTT). The participants were classified into four groups: normal glucose tolerance (NGT), NGT-hyperinsulinemia (NGT-HI), impaired glucose tolerance (IGT), and diabetes mellitus (DM). Blood chemistries, including FPG, glycated hemoglobin, and lipid profiles, and liver function test were performed every 6–12 months or when the patient developed any symptom or sign indicative of diabetes.
Results: Glucose metabolism alterations were detected in 81.4% of the participants: 63.8% with NGT-HI, 15.3% with IGT, and 2.3% with T2DM. The median levels of homeostasis model assessment-insulin resistance (HOMA-IR) in patients with IGT (8.63) were significantly greater than those in the patients with NGT (4.04) ($p < 0.01$). During the follow-up, 22 patients (14.4%) developed T2DM significantly more from the IGT group (nine of 33 cases, 27.3%) than the NGT-HI group (12 of 108 cases, 11.1%) ($p = 0.022$). The predicting parameters for T2DM conversion were weight status, body mass index (BMI), FBG, fasting insulin, alanine transaminase (ALT) levels, and HOMA-IR.
Conclusions: Glucose metabolism alteration was commonly found among obese adolescents. Factors associated with T2DM development were greater weight status and the severity of insulin resistance as shown by higher HOMA-IR levels.

Keywords: impaired glucose tolerance; insulin resistance; obesity; type 2 diabetes mellitus.

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Introduction

The prevalence of obesity in children and adolescents is increasing worldwide [1, 2]. In Thailand, the prevalence of obesity in school-aged children increased from 5.8% in 1995 to 9.7% in 2009 [3], and in adolescents 18%–20% in 2008–2009 [4]. It is known that obesity is associated with many health problems, particularly abnormal glucose metabolism, dyslipidemia, hypertension, and non-alcoholic fatty liver disease (NAFLD) [5, 6]. The comorbidities of abnormal glucose metabolism include insulin resistance or hyperinsulinemia (HI), impaired glucose tolerance (IGT), and type 2 diabetes mellitus (T2DM), which can be detected through glucose loading by the oral glucose tolerance test (OGTT) [7]. Studies regarding the prevalence of abnormal glucose metabolism in obese adolescents have shown various results from various study populations, for example 7%–20% of overweight/obese Caucasian adolescents had IGT [8–10] and 3.3% of overweight/obese Chinese children and adolescents had IGT [11].

In Thailand, cross-sectional studies in obese children and adolescents have found high prevalence of impaired glucose metabolism as revealed by an OGTT of 25% [12, 13]. However, there has been no longitudinal study of obese youth comparing those with and without baseline abnormal glucose metabolism. In our Pediatric Endocrine Clinic at Songklanagarind Hospital in southern Thailand,

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we performed an OGTT in 177 obese children and adolescents and followed them up for 3–6 years. The purpose of this study was to evaluate the prevalence of abnormal glucose metabolism in asymptomatic severely obese children and adolescents, and the association between abnormal glucose tolerance and biochemical markers or insulin dynamic markers and to determine the percentage of these children who later developed diabetes and categorizing the main predictors for diabetes development in Thai youth with severe obesity.

Materials and methods

This was an observational cohort study conducted from 2007 to 2013 on obese Thai children and adolescents, aged 8–15 years, as defined by the criterion of body mass index (BMI) over the 95th percentile for age [14]. During the study period, 181 obese children and adolescents were identified. Of these, four were excluded due to receiving treatment with metformin for impaired fasting glucose (IFG) as defined by fasting plasma glucose (FPG) between 105 and 120 mg/dL. Hence, 177 obese children and adolescents (93 boys, 52.5%), aged 8–15 years, underwent an OGTT at the Pediatric Endocrine Clinic at Songklanagarind Hospital. All patients had a previous plasma glucose level of < 100 mg/dL, had no clinical findings of secondary obesity which can be caused by syndromes, such as Prader-Willi syndrome, etc., and were not on corticosteroids for treatment of an unrelated disease such as a connective tissue disease, allergic disease, etc.

Data collection at the time of recruitment included related history (age at onset of obesity, family history of obesity, and/or type 2 diabetes) and a physical examination [weight and height measured by a standard method, presence of acanthosis nigricans, blood pressure, and Tanner stage of breast (in girls) or testicular volume (in boys) evaluated according to the Marshall and Tanner method]. To account for age and sex discrepancies, weight and height in kg and cm, respectively, were transformed to standard deviation scores (SDSs) based on chronological age using the standardized reference data of Thai children [15]. BMI was calculated by weight in kilograms divided by the square of height in meters, and then transformed to SDS according to the World Health Organization (WHO) database [14]. Laboratory investigations included fasting serum cholesterol, triglyceride, high-density lipoprotein-cholesterol (HDL-C), low-density lipoprotein-cholesterol (LDL-C), alanine transaminase (ALT), aspartate transaminase (AST), and hemoglobin A_{1c} (HbA_{1c}). Plasma glucose levels were measured by the hexokinase enzymatic method and HbA_{1c} levels by the turbidimetric inhibition immunoassay. Serum insulin levels were measured by electrochemiluminescence immunoassay (ECLIA) (Roche Diagnostics, IN, USA). Serum cholesterol levels were measured by the cholesterol oxidase/peroxidase aminophenazone (CHOD-PAP) reagent, triglyceride levels by the glycerol-3-phosphate oxidase (GPO)/PAP enzymatic colorimetric method, HDL-C levels by the polyethylene glycol (PEG)-modified enzymes, LDL-C levels by a direct method, and alanine aminotransferase and aspartate aminotransferase levels by the International Federation of Clinical Chemistry (IFCC) method.

OGTT and interpretation

OGTTs were performed after at least 8 h of overnight fasting. The child was given a standard 1.75 g/kg (maximum 75 g) glucose solution load, and venous blood specimens were collected at 0, 30, 60, 90, and 120 min to measure plasma glucose and insulin levels.

The definitions and classification of diabetes types according to the OGTT were based on the 1999 WHO criteria and the 2011 ISPAD American Diabetes Association criteria [16]. In this study, the participants were classified into four groups according to the results of their OGTT as follows:

- Group 1: Normal glucose tolerance without insulin resistance (NGT): FPG < 101 mg/dL, 2-h glucose level from the OGTT < 140 mg/dL, and normal levels of serum insulin.
- Group 2: Normal glucose tolerance with hyperinsulinemia (NGT-HI): FPG < 101 mg/dL, 2-h OGTT glucose level < 140 mg/dL, and elevated levels of serum insulin in either fasting > 15 mIU/mL or peak level > 150 mIU/mL or at 120 min of > 75 mIU/mL [17].
- Group 3: IGT: FPG < 101 mg/dL, and 2-h OGTT glucose level ≥ 140 and < 200 mg/dL.
- Group 4: Diabetes mellitus (DM): FPG > 126 mg/dL and 2-h OGTT glucose level ≥ 200 mg/dL.

Insulin and beta-cell function indices

Whole body insulin sensitivity index (WBISI) [18] and homeostasis model assessment-insulin resistance (HOMA-IR) levels [19] were calculated for assessment of approximation of whole body insulin sensitivity and hepatic insulin insensitivity, respectively. A fasting glucose insulin ratio (FGIR) was calculated to assess for insulin resistance [20]. Insulinogenic index (IGI) and HOMA-β levels were calculated for beta-cell function and insulin secretion, respectively [21]. These indices were calculated using the following equations:

$$\text{WBISI} = 10000 / \sqrt{(\text{FPG (mg/dL)} \times \text{FI } (\mu\text{U/mL}) \times \text{mean G (mg/dL)} \times \text{mean I } (\mu\text{U/mL})},$$

$$\text{HOMA-IR} = [\text{FI } (\mu\text{U/mL}) \times \text{FPG (mmol/L)}]/22.5,$$

$$\text{FGIR} = \text{FPG (mg/dL)}/\text{FI } (\mu\text{U/mL}),$$

$$\text{IGI} = \Delta \text{ insulin } 0\text{–}30 \ (\mu\text{U/mL})/\Delta \text{ glucose } 0\text{–}30 \ (\text{mg/dL}),$$

$$\text{HOMA-}\beta = [20 \times \text{FI } (\mu\text{U/mL})]/[\text{FPG (mmol/L)} - 3.5],$$

where FPG indicates fasting plasma glucose, FI indicates fasting plasma insulin, mean G indicates mean plasma glucose, mean I indicates mean plasma insulin, and Δ insulin 0–30 and Δ glucose 0–30 indicate the increments of insulin and glucose at 30 min of the OGTT, respectively.

Follow-up of patients

Following their initial assessment, each patient was regularly followed up every 3–6 months and weight and height measured, and advice given concerning behavioral intervention for weight reduction. Blood chemistries, including FPG, HbA_{1c}, and lipid profiles, and liver function test were performed every 6–12 months or when

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the patient developed any symptoms or signs indicative of diabetes such as not feeling well, polyuria, polydipsia, nocturia, or weight loss.

Statistical analysis

Data are expressed as median with interquartile range (IQR) for continuous numbers and percentage for categorical numbers. Chi-squared test, analysis of variance (ANOVA), and Wilcoxon-ranked sum test were used to compare the differences in categorical data. Survival analysis was used for the average time for T2DM development during the follow-ups. Cox regression analysis for receiver operating characteristic (ROC) curves was used to analyze the reasonable cut-off values of selected indices for predicting diabetes development. All data analyses were performed using R program (R Foundation, Austria, available from http://www.r-project.org/foundation/main.html). Statistical differences were considered significant at a p-value of < 0.05.

Ethical approval

The protocol for this study was approved by the Institution Review Board and the Ethics Committee of the Faculty of Medicine, Prince of Songkla University. Written informed consent was obtained from all participants and their parents for permission to use the clinical data for this study.

Results

Patient characteristics

The median age of the subjects was 12.2 years (IQR, 9–14), with 58 prepubertal (32.8%). The median BMI was 33.4 kg/m^2 (IQR, 28.4–56.8). The characteristics of all participants are shown in Table 1.

OGTT and glucose tolerance status

According to the glucose metabolism status as revealed by the OGTT, the patients were divided into four groups: NGT (n = 33, 18.6%), NGT-HI (n = 113, 63.8%), IGT (n = 27, 15.3%), and DM (n = 4, 2.3%) as diabetes. The median levels and IQRs of plasma glucose and serum insulin by the OGTT are shown in Figure 1. Plasma glucose levels at

Table 1: Characteristics of study population (n = 177).

	Median (IQR)	X ± SD
Age, years	12.2 (9–14)	11.9 ± 2.5
Weight, kg	81.4 (52.5–148.3)	83.8 ± 22.5
Weight SDS	5.81 (2.53–17.03)	6.26 ± 2.61
Height, cm	155 (130.3–175)	154.6 ± 13.5
Height SDS	1.30 (1.33–3.62)	1.36 ± 1.43
Body mass index, kg/m^2	33.4 (28.4–56.8)	34.7 ± 6.5
Weight-for-height, %	180.9 (145.7–364.8)	188.5 ± 40.7
Tanner stage, n (%)		
I	58 (32.8)	–
II–III	55 (31.1)	–
IV–V	64 (36.1)	–
DM in family members, n (%)	78 (44.1)	–
First degree relatives, n (%)	44 (24.9)	–
Second degree relatives, n (%)	34 (19.2)	–
Obesity in family members, n (%)	101 (57.1)	–
First degree relatives, n (%)	88 (49.7)	–
Second degree relatives, n (%)	13 (7.3)	–
Systolic blood pressure, mmHg	123 (90–162)	123 ± 14
Diastolic blood pressure, mmHg	74 (50–95)	74 ± 9
AST, IU/L	26.5 (13–197)	36.8 ± 28.8
ALT, IU/L	35 (10–300)	49.6 ± 44.4
Cholesterol, mg/dL	190 (92–323)	190.9 ± 36.2
Triglyceride, mg/dL	125.5 (48–403)	132.0 ± 55.0
HDL-C, mg/dL	43.1 (27.9–72.6)	44.8 ± 8.6
LDL-C, mg/dL	129.5 (23.7–245.6)	128.4 ± 32.6

SDS, standard deviation score; ALT, alanine transaminase; AST, aspartate transaminase; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol.

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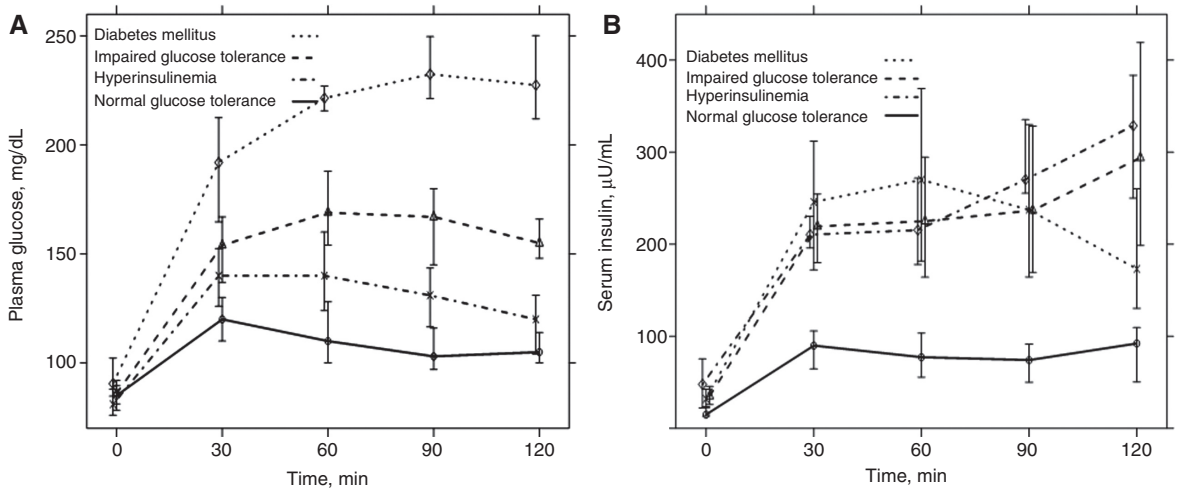

Figure 1: Median levels and interquartile ranges of plasma glucose (A) and serum insulin (B) after oral glucose tolerance test.

baseline and at each time after the OGTT of the patients in IGT group were significantly greater than those in the NGT-HI and NGT groups ($p < 0.001$). Serum insulin levels were significantly greater in NGT-HI and IGT groups at 30, 60, 90, and 120 min after the OGTT than those in the NGT group ($p < 0.01$). It was of note that the plasma glucose and insulin levels increased with the severity of abnormal glucose tolerance.

The clinical characteristics of the participants in the NGT, NGT-HI, and IGT groups are shown in Table 2. The weight and height of participants in the NGT-HI and IGT groups were significantly greater than those in the NGT group; however, when adjusted through the SDSs, there were no differences in weight and height SDSs among the three groups. The median HOMA-IR levels of the participants in the NGT-HI and IGT groups were significantly greater than those in the NGT group ($p < 0.01$), and these levels also increased with the severity of abnormal glucose metabolism. The median FGIR and WBISI levels, however, were significantly lower in the NGT-HI and IGT groups than those in the NGT group ($p < 0.01$). The median IGI and HOMA-β levels were significantly greater in participants in the NGT-HI and IGT groups than those in the NGT group.

Longitudinal follow-up

During the 3–6-year follow-up period, 20 participants (11.6%) were lost to follow-up, five from the NGT group, and 15 from the NGT-HI group. Of the 153 participants, 22 (14.4%) had a FPG level over 126 mg/dL, later confirmed by the average HbA_{1c} level of 7.71% ± 1.45% (range 6.6%–13.3%). Those who were not diagnosed as DM had an average HbA_{1c} level of 5.5% ± 0.4% (range 4.7%–5.9%). Most of the patients who developed T2DM were from the IGT and NGT-HI groups ($p = 0.022$): nine of 33 patients (27.3%) from the IGT group, 12 of 108 patients (11.1%) from the NGT-HI group, and one of 28 patients (3.6%) from the NGT group. The median durations from the time the OGTT was performed until the diagnosis of T2DM were significantly shorter in the IGT group (1.70 years, IQR: 1.5–3.2) and the NGT-HI group (2.45 years, IQR: 1.9–3.9) than those in the NGT group (5.9 years) ($p < 0.01$) (Figure 2). The clinical characteristics and laboratory results of the participants who later developed T2DM and those who were still non-diabetic at the end of this study period are compared in Table 3. The characteristics with significant differences between these two groups were weight and weight SDS, and BMI and BMI SDS, both at the time the OGTT was performed and at the time of follow-up ($p < 0.01$); weight gain was also significantly greater in these groups during the follow-up period ($p < 0.01$). The significantly different laboratory parameters were FPG, fasting serum insulin, peak plasma glucose, AST and ALT levels, HOMA-IR, and WBISI.

Prediction for T2DM development

ROC curves were used to analyze reasonable cut-off values for prediction of T2DM development of the selected significant indices of FPG, fasting serum insulin, HOMA-IR, and WBISI (Figure 3). The greatest area under the curve was FPG at 77.1% (95% CI: 64.6%–89.6%), followed by HOMA-IR at 69.6% (95% CI: 56.5%–82.8%), fasting serum insulin at 65.0% (95% CI: 52.4%–77.7%), and WBISI at 67.4% (95% CI: 53.2%–81.5%). The most suitable cut-off levels for prediction of T2DM development were FPG of 88 mg/dL (sensitivity 80%, specificity 68%), fasting

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Table 2: Characteristics of study subjects in each category of glucose metabolism.

Predictor	NGT (n = 33)	NGT-HI (n = 113)	IGT (n = 27)	p-Value
Age, years	11.7 (10.7 ± 3.1)	12.4 (12.1 ± 2.4)	12.6 (12.4 ± 2.3)	0.02
Gender, male, n (%)	17 (51.5)	59 (52.0)	17 (63.0)	0.15
Weight, kg	71.9 (71.6 ± 19.9)	85.8 (86.6 ± 22.8)	87.7 (86.7 ± 19.7)	0.007
Weight SDS	5.65 (5.66 ± 2.92)	6.11 (6.41 ± 2.43)	6.17 (6.37 ± 2.21)	0.37
Height, cm	145.8 (145.8 ± 14.3)	156.4 (156.4 ± 13.7)	155.9 (156 ± 10)	0.01
Height SDS	1.14 (0.94 ± 1.39)	1.32 (1.52 ± 1.29)	1.28 (1.28 ± 1.47)	0.12
BMI, kg/m^2	32.5 (33.5 ± 7.2)	34.0 (35.0 ± 6.3)	34.3 (35.3 ± 5.8)	0.51
BMI SDS	3.52 (3.67 ± 1.09)	3.51 (3.67 ± 0.95)	3.51 (3.66 ± 0.86)	0.75
Tanner stage, n (%)				0.78
I	10 (30.3)	36 (31.9)	8 (29.6)	
II–III	11 (33.3)	36 (31.9)	8 (29.6)	
IV–V	12 (36.4)	41 (36.2)	11 (40.8)	
Age at obesity, years	4.5 (4.8 ± 2.5)	4.5 (5.5 ± 2.8)	4.2 (6.3 ± 2.5)	0.58
SBP, mmHg	115 (119 ± 12)	121 (124 ± 14)	121 (124 ± 15)	0.25
DBP, mmHg	72 (74 ± 10)	74 (74 ± 9)	76 (77 ± 9)	0.34
AST, IU/L	29 (29 ± 9)	35 (35 ± 24)	38 (39 ± 29)	0.28
ALT, IU/L	33 (34 ± 19)	47 (49 ± 41)	51 (52 ± 41)	0.17
Cholesterol, mg/dL	178 (191 ± 40)	182 (189 ± 34)	184 (188 ± 31)	0.74
Triglyceride, mg/dL	115 (115 ± 45)	131 (131 ± 49)	135 (136 ± 55)	0.32
HDL-C, mg/dL	47 (47 ± 10)	44 (44 ± 8)	44 (44 ± 8)	0.32
LDL-C, mg/dL	120 (130 ± 37)	127 (127 ± 32)	125 (125 ± 26)	0.89
Peak glucose, mg/dL	125 (125.3 ± 14.4)	149 (149.8 ± 23.9)	180.8 (180.8 ± 27.9)	< 0.001
Peak insulin, μU/mL	109.4 (109.4 ± 42.4)	351.9 (351.9 ± 179.5)	381.0 (381.0 ± 208.2)	< 0.001
HOMA-IR	3.23 (4.04 ± 3.17)	6.37 (7.20 ± 4.29)	7.10 (8.63 ± 4.76)	< 0.001
FGIR	5.33 (5.27 ± 1.88)	2.73 (2.92 ± 1.42)	2.40 (2.69 ± 1.06)	< 0.001
WBISI	3.48 (3.32 ± 0.99)	1.28 (1.44 ± 0.64)	0.97 (1.10 ± 0.41)	< 0.01
IGI	2.32 (2.28 ± 1.3)	3.95 (4.22 ± 2.22)	2.68 (3.16 ± 1.79)	< 0.01
HOMA-β	288 (377 ± 273)	613 (775 ± 544)	563 (595 ± 252)	< 0.01

Data points are shown as median (mean ± standard deviation) (n = 173; four participants with diabetes were excluded). X ± SD, mean ± standard deviation; SDS, standard deviation score; BMI, body mass index; WFH, weight-for-height; SBP, systolic blood pressure; DBP, diastolic blood pressure; ALT, alanine transaminase; AST, aspartate transaminase; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; HOMA-IR, homeostasis model assessment-insulin resistance; FGIR, fasting glucose insulin ratio; IGI, insulinogenic index; WBISI, whole body insulin sensitivity index; HOMA-β, homeostasis model assessment-β.

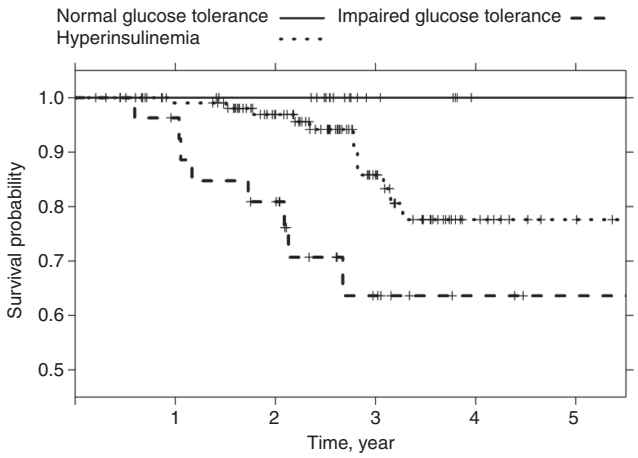

Figure 2: Median durations from the time of the OGTT was performed until the time of diagnosis of T2DM.

serum insulin of 30 μU/mL (sensitivity 80%, specificity 56%), HOMA-IR of 7.1 (sensitivity 75%, specificity 72%), and WBISI of 1.0 (sensitivity 64%, specificity 76%).

Discussion

The OGTTs in our obese children and adolescents revealed that 81.4% had abnormal glucose metabolism: 63.8% with NGT-HI, 15.3% with IGT, and 2.3% were silent asymptomatic T2DM. The high percentage of abnormal glucose metabolism detected by the OGTT in our obese children and adolescents was similar to other previous studies in obese Thai children and adolescents [12, 13, 22, 23], and studies from other countries [6, 8–11]. Recent studies have found that prevalence of IGT and T2DM has increased along with increased prevalence of obesity, mostly during adolescence, and T2DM has become a major concern in obese children and adolescents [6–9, 24, 25]. To our knowledge, there have been no longitudinal studies examining the conversion rate from IGT to T2DM in obese Thai or other Asian children. Our study in obese adolescents found a high conversion rate of 27% from IGT to T2DM within the average time of 1.7 years and 10% from NGT-HI

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Table 3: Comparison of clinical characteristics and laboratory parameters of participants who later developed diabetes (n = 22) and those who were still non-diabetic at the follow-up (n = 131).

Predictor	DM (n = 22)	Non-DM (n = 131)	p-Value
Gender, male, n (%)	14 (63.6)	69 (52.7)	0.34
Age OGTT done, years	12.7 (12.3 ± 2.7)	11.9 (11.8 ± 2.5)	0.09
Weight, kg	89.1 (94.6 ± 26.3)	79.5 (80.5 ± 20.1)	< 0.001
Weight SDS	6.68 (7.74 ± 3.57)	5.63 (5.80 ± 2.04)	< 0.001
Height, cm	155.0 (156.5 ± 14.3)	155.5 (154.0 ± 13.3)	0.31
Height SDS	1.50 (1.46 ± 1.91)	1.44 (1.32 ± 1.26)	0.59
BMI, kg/m^2	36.6 (38.3 ± 8.30)	32.9 (33.5 ± 5.4)	< 0.001
BMI SDS	3.95 (4.21 ± 1.27)	3.59 (3.69 ± 0.88)	< 0.001
Age at obesity, years	5.0 (5.5 ± 2.7)	5.0 (5.5 ± 2.7)	0.99
History of DM in parents, n (%)	8 (36.4)	24 (18.3)	0.054
Obesity in parents, n (%)	12 (54.5)	50 (38.2)	0.15
SBP, mmHg	123 (123 ± 17)	123 (123 ± 13)	0.99
DBP, mmHg	72 (71.7 ± 9.0)	75 (74.6 ± 8.8)	0.27
AST, IU/L	37.5 (54.7 ± 43.1)	25 (31.8 ± 20.9)	< 0.001
ALT, IU/L	53.5 (72.2 ± 58.5)	33.0 (43.0 ± 37.1)	< 0.001
Non-alcoholic fatty liver disease, n (%)	13 (59.1)	34 (26.0)	0.002
Cholesterol, mg/dL	197 (193.4 ± 33.1)	188 (190.0 ± 37.1)	0.60
Triglyceride, mg/dL	129 (138.0 ± 70.0)	125 (130.1 ± 49.7)	0.42
HDL-C, mg/dL	44.9 (45.7 ± 8.3)	42.6 (44.5 ± 8.7)	0.43
LDL-C (mg/dL)	130 (126.5 ± 29.7)	129 (129.0 ± 33.6)	0.67
Fasting plasma glucose, mg/dL	91 (92.8 ± 14.4)	81 (82.5 ± 7.0)	< 0.01
Fasting serum insulin, μU/mL	37.5 (38.9 ± 18.9)	26.6 (32.3 ± 19.5)	0.03
Peak glucose, mg/dL	180 (182.8 ± 40.2)	142 (145.5 ± 23.6)	< 0.001
Peak insulin, μU/mL	379 (356 ± 133)	261 (304 ± 194)	0.15
HOMA-IR	8.72 (9.04 ± 5.20)	5.64 (6.64 ± 4.19)	0.004
FGIR	2.69 (2.98 ± 1.92)	3.04 (3.28 ± 1.65)	0.17
WBISI	0.95 (1.32 ± 0.93)	1.42 (1.75 ± 1.01)	0.02
IGI	2.84 (3.63 ± 2.23)	3.46 (3.76 ± 2.13)	0.72
HOMA-β	456 (589 ± 413)	561 (606 ± 502)	0.28
Sum insulin, μU/mL	1022 (1034 ± 475)	860 (971 ± 604)	0.33
AUC insulin	2657 (2728 ± 1322)	2155 (2547 ± 1696)	0.35
At follow-up			
Duration of follow-up, years	2.4 (2.3 ± 1.5)	2.4 (3.1 ± 1.5)	0.08
Age at last visit, years	14.7 (15.0 ± 3.0)	14.8 (14.9 ± 2.8)	0.65
Weight, kg	99.5 (98.8 ± 20.4)	86.0 (87.5 ± 18.7)	< 0.001
Weight SDS	7.54 (8.44 ± 3.20)	5.62 (5.85 ± 1.98)	< 0.001
Weight gain, kg	8.7 (8.4 ± 4.3)	5.4 (5.5 ± 2.8)	< 0.001
Height, cm	161.5 (161.4 ± 9.7)	158.0 (158.5 ± 11.5)	0.24
Height SDS	1.40 (1.46 ± 1.20)	1.42 (1.44 ± 1.15)	0.75
BMI, kg/m^2	37.9 (38.5 ± 8.5)	33.5 (33.6 ± 6.2)	< 0.001
BMI SDS	3.58 (3.66 ± 1.18)	2.96 (2.97 ± 0.86)	< 0.001
HbA_{1c} at last visit, %	7.20 (7.71 ± 1.45)	5.60 (5.50 ± 0.5)	< 0.001

Data points are shown as median (mean ± standard deviation). X ± SD, mean ± standard deviation; SDS, standard deviation score; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; ALT, alanine transaminase; AST, aspartate transaminase; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; HOMA-IR, homeostasis model assessment-insulin resistance; FGIR, fasting glucose insulin ratio; IGI, insulinogenic index; WBISI, whole body insulin sensitivity index; HOMA-β, homeostasis model assessment-β; HbA_{1c}, hemoglobin A_{1c}.

to T2DM within 2.5 years after the detection of abnormal glucose metabolism. The probability of such conversion was related to weight status, particularly weight gain and BMI, severity of insulin resistance, and the presence of NAFLD.

The current increasing rate of T2DM in the adolescent age group is related not only to the increasing prevalence of obesity, but also associated with various other factors in the modern world such as unhealthy eating behaviors and increasingly sedentary lifestyles [1, 2]. Moreover, during

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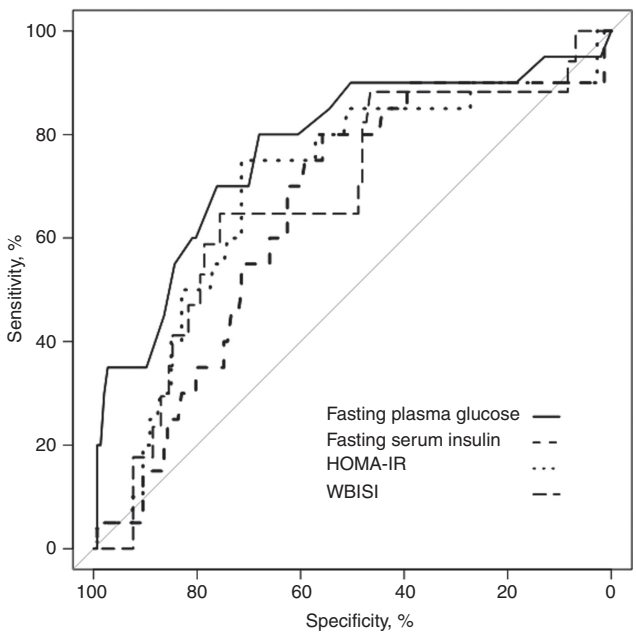

Figure 3: ROC curves for cut-off values for prediction of T2DM development of the selected significant indices.

adolescence, physiological insulin resistance occurs as a result of the increase in growth hormone and sex steroid hormone secretions during puberty [26, 27]. Hence, physiological hormonal changes during puberty may exacerbate insulin resistance in obese adolescents whose obesity started during childhood, and the increased insulin resistance then increases the risk of progression from NGT-HI to IGT to T2DM. Recent studies in obese adolescents, mostly from western countries, have shown different conversion rates from IGT to T2DM of 2%–30% within 2–5 years [28–30] and even within 1 year [31]. However, about 70%–80% of the obese adolescents in those studies turned out to be NGT. The differences in conversion rates were postulated to be from lifestyle modification and weight status. However, a high conversion rate from IGT to NGT in those studies should be interpreted with caution as most of the children improved their weight status which might have been related to the good outcomes. Moreover, analyzing only IGT children with available follow-up cases in those published studies might overestimate the conversion from IGT to NGT, as children who are motivated to change their lifestyle behaviors will likely also present more frequently at follow-up. In our study, we had no data concerning the conversion from IGT to NGT because an OGTT was performed only once at the initial evaluation. However, we found a high conversion rate in our obese adolescents, of whom 27% with IGT developed T2DM within 1.7 years of the original test, particularly IGT patients who experienced greater weight gain than those who did not develop T2DM during the follow-up time period.

Our study found that all of the dynamic insulin indices were significantly different among the three groups by OGTT classification and related to the severity of abnormal glucose metabolism. However, after 3–6 years of follow-up, only HOMA-IR and WBISI were significantly different between those who later developed T2DM and those who did not. The high HOMA-IR and low WBISI levels indicate that higher hepatic insulin insensitivity and lower whole body insulin sensitivity in severely obese youth result in greater deterioration of glucose metabolism and could be used as parameters for predicting later T2DM development. A study in American adolescents (12–19 years) found that the 95th percentile HOMA-IR level was 4.4, and an HOMA-IR level > 4.5 was the most practical definition of insulin resistance in adolescents [2, 24]. A recent literature review found that obese children in south Asian countries had a greater tendency to develop insulin resistance at an early age than White Caucasian children [32]. The results of our study confirmed a high prevalence of insulin resistance in obese children and adolescents, as shown by the high HOMA-IR levels even in the NGT group, and also higher HOMA-IR levels in Thailand than the results reported in Caucasian countries. Hence, we conclude that HOMA-IR is a suitable and practical test for detection of abnormal glucose metabolism in a clinical setting as it requires only a single fasting blood test. If an HOMA-IR level is > 7.0, then an OGTT should be performed to identify children at risk for IGT and T2DM.

In our study, the significant differences in clinical parameters which might identify obese youths at risk to develop T2DM from obese adolescents who retained normal glucose metabolism were BMI > 32 kg/m^2, FPG > 88 mg/dL, fasting insulin > 30 mIU/L, and HOMA-IR > 7.0. Elevated ALT levels of 70 IU/L or twice the normal ranges were also significantly related to a high rate of conversion from IGT to T2DM, a finding which is consistent with a recent study in Caucasian youth which found that raised ALT was the hepatic manifestation of insulin resistance [2]. The cut-off FPG threshold of 88 mg/dL as an indicator for obese children at risk to develop T2DM in our study is consistent with the cut-off FPG level of 86 mg/dL reported by Maffeis et al. [33] and Grandone et al. [34] Although a FPG level of 88 mg/dL is within the normal ranges, it is also relatively high in patients with insulin resistance.

This current study has a notable strength and some limitations. The strength is the longitudinal follow-up for 3–6 years with only 18% of participants lost to follow-up. The limitations are first that it was a single hospital-based study in severely obese youth, and that it did not include mildly obese youth, thereby possibly overestimating the frequency of IGT and T2DM. Second, we did not include

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the IFG patients who were treated with metformin as treatment might modify the natural course of abnormal glucose metabolism during follow-up. Third, we did not include information on physical activity or the caloric food intake of our participants because of poor compliance and unreliable daily records. Finally, our study did not recruit normal weight children and adolescents in a control group as it would be inappropriate to perform OGTTs with multiple blood samples in normal weight children and adolescents.

In summary, our study found a high rate of abnormal glucose metabolism in obese children and adolescents. A high conversion rate from abnormal glucose metabolism to T2DM was found in obese children and adolescents with IGT. The main factors associated with T2DM development in these adolescents were greater weight status, high HOMA-IR level, and the presence of NAFLD. Obese children who are at high risk of abnormal glucose metabolism should be identified and appropriate intervention should be undertaken prior to development of T2DM. Prompt intervention to deal with developing childhood obesity should also be undertaken to decrease the associated development of impaired glucose metabolism and T2DM development in later adolescence.

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